CT scan reconstruction with deep learning

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1 Generating a phantom and a sinogram

- Generate a phantom where each pixel consists of part bone and part water described as (bone, water)
  - Background values should be low for both to simulate air (0, 0.05)
  - main body should be elliptical and mainly consist of water (0.20, 0.80)
  - elliptical inserts should simulate different organs and/or body featurs (0.80, 0.80)
- Generate a sinogram from the projections
  - The function will return a 3D object containing the sinograms for the height of the detector. We want to select a slice to display
  - print a 2D cross section of the phantom with a corresponding sinogram

```
#settings
objectSize = 32     # size of the object
nPixelsY = 64       # number of pixels in Y direction detector
nPixelsZ = 44       # number of pixels in Z direction detector
projections = 64    # number of projections
iterations = 40     # number of iterations
```

1.0.1 generate phantom

```
import random, numpy as np

def generate_phantom(n, h, num_features=1):

    h = random.randint(1, 3)  # Random height offset for the torso

    # Initialize 3D bone and water matrices with zeros
    bone_matrix = np.zeros((n, n, n), dtype=float)
    water_matrix = np.zeros((n, n, n), dtype=float)

    # fill background of matrices with 0.05 bone and water
    bone_matrix.fill(0.05)
    water_matrix.fill(0.05)

    # set values for main ellipse (simulated torso)
    intensity_bone = random.uniform(0.1, 0.6)
    intensity_water = random.uniform(0.1, 0.6)  # Random intensity for water

    offset_i = np.random.randint( -h, h + 1)
    offset_j = np.random.randint( -h, h + 1)

    center_i = n // 2 + offset_i
    center_j = n // 2 + offset_j

    # Main body ellipse (simulated torso with random offset)
    for i in range(h, n - h):
        for j in range(2 * h, n - 2 * h):
            if ((i - center_i) / ((n - 2 * h) / 2)) ** 2 + ((j - center_j) / ((n - 6 * h) / 2)) ** 2 <=
                bone_matrix[2*h: -2*h, i, j] = intensity_bone
                water_matrix[2*h: -2*h, i, j] = intensity_water

    # Adding smaller features (e.g., organs or bone structures)
    for _ in range(num_features):
```

```
        # Random place in the body - h is the offset from the edge, n is the size of the matrix
        a = random.randint(h, (n - h) // 4)  # Semi -major axis
        b = random.randint(h, (n - h) // 4)  # Semi -minor axis
        center_x = random.randint(h + a, n - h - a)
        center_y = random.randint(3 * h + b, n - 3 * h - b)

        a_water = random.randint(h, (n - h) // 4)  # Semi -major axis for water
        b_water = random.randint(h, (n - h) // 4)  # Semi -minor axis for water

        # Randomly choose the intensity for bone and water, together they add to 1.
        bone_intensity = 0.8
        water_intensity = 0.8

        for i in range(center_x - a_water, center_x + a_water):
            for j in range(center_y - b_water, center_y + b_water):
                if 0 <= i < n and 0 <= j < n:
                    if ((i - center_x) / a_water) ** 2 + ((j - center_y) / b_water) ** 2 <= 1:
                        bone_matrix[2*h: -2*h, i, j] = bone_intensity

        center_x = random.randint(h + a, n - h - a)
        center_y = random.randint(3 * h + b, n - 3 * h - b)

        for i in range(center_x - a, center_x + a):
            for j in range(center_y - b, center_y + b):
                if 0 <= i < n and 0 <= j < n:
                    if ((i - center_x) / a) ** 2 + ((j - center_y) / b) ** 2 <= 1:
                        water_matrix[2*h: -2*h, i, j] = water_intensity

    water_matrix *= 1  # Water density is 1
    bone_matrix *= 1.92  # Apply bone density

    return bone_matrix, water_matrix

bone, water = generate_phantom(objectSize, 2, num_features=1)
x = np.column_stack((bone.ravel(), water.ravel()))
```

1.0.2 Running projection algorithm

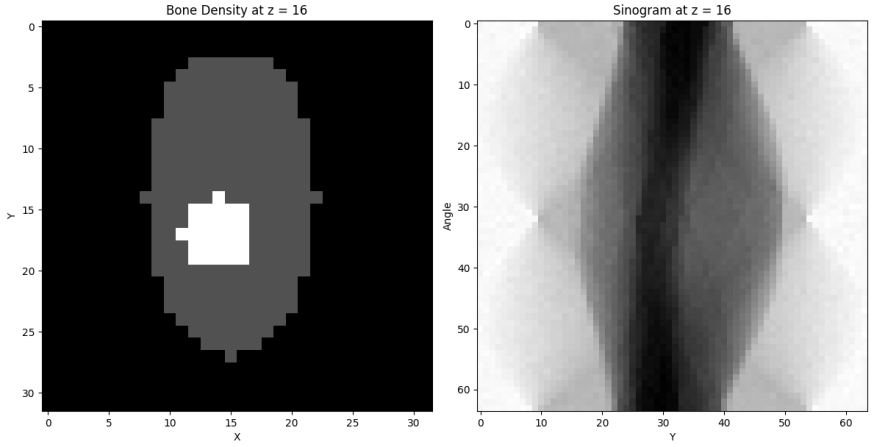
```
from libs.simulatepreps import projectMatrix

#generate proection matrices
y, S, M, A = projectMatrix(x, objectSize, nPixelsY, nPixelsZ, 1, projections)
```

1.0.3 Generating sinogram

```
def getSinogram(y, energy_bin_index=0, z_index=32, nProj=100, detector_shape=(64, 44)):
    nX, nZ = detector_shape
    y_bin = y[energy_bin_index]
    y_proj_zx = y_bin.reshape((nProj, nZ, nX))  # shape: (100, 64, 44)
    sinogram = y_proj_zx[:, z_index, :]  # shape: (100, 44)
    return sinogram

sinogram = getSinogram(y, energy_bin_index=0, z_index=objectSize // 2, nProj=projections, detector_shape
```

1.0.4 Make figure

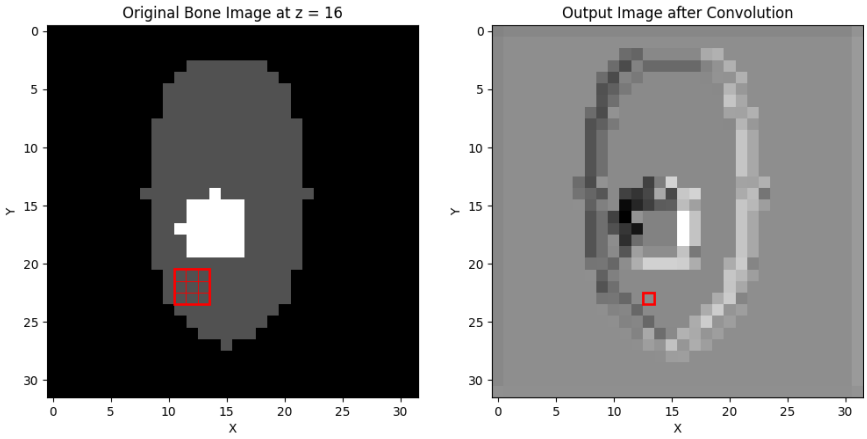
```
import matplotlib.pyplot as plt


# plot bone density and phantom at objectSize //2
plt.figure(figsize=(12, 6))
plt.subplot(1, 2, 1)
plt.imshow(bone[objectSize // 2, :, :], cmap='gray')
plt.title('Bone Density at z = {}'.format(objectSize // 2))
plt.xlabel('X')
plt.ylabel('Y')

plt.subplot(1, 2, 2)
plt.imshow(sinogram, cmap='gray')
plt.title('Sinogram at z = {}'.format(objectSize // 2))
plt.ylabel('Angle')
plt.xlabel('Y')

plt.tight_layout()
plt.show()
```



1.1 Visualising a 2D convolution

```
# we want to put the phantom through a pytorch 2d convolution layer
import torch

bone_array = bone[objectSize // 2, :, :]
bone_tensor = torch.tensor(bone[objectSize // 2, :, :], dtype=torch.float32)

# create a 2D convolution layer with a kernel size of 3x3
conv_layer = torch.nn.Conv2d(in_channels=1, out_channels=1, kernel_size=3, padding=1)

# apply the convolution layer to the bone tensor
output_tensor = conv_layer(bone_tensor.unsqueeze(0).unsqueeze(0))  # Add batch and channel dimensions
output_image = output_tensor.squeeze(0).squeeze(0).detach().numpy()  # Remove batch and channel dimensi

# plot 2 plots, one of the original bone image with a square showing one of the convolution kernels, an
plt.figure(figsize=(12, 6))
```

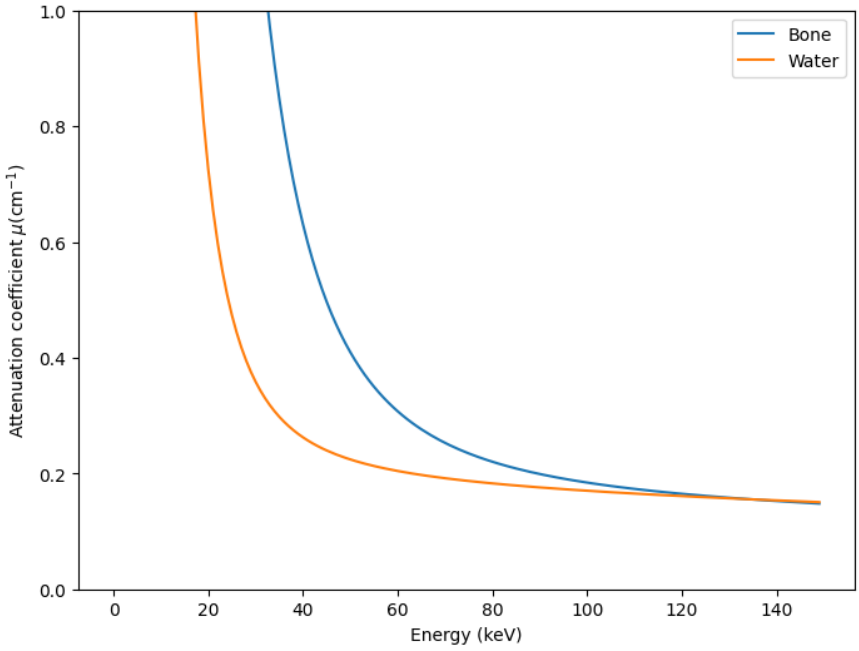
```
plt.subplot(1, 2, 1)
plt.imshow(bone_tensor.numpy(), cmap='gray')
# draw a square representing the convolution kernel
kernel_size = conv_layer.kernel_size[0]
plt.gca().add_patch(plt.Rectangle((10.5, 20.5), kernel_size, kernel_size, edgecolor='red', facecolor='n
# fill square with rectengles for each pixel
for i in range(kernel_size):
    for j in range(kernel_size):
        plt.gca().add_patch(plt.Rectangle((10.5 + i, 20.5 + j), 1, 1, edgecolor='red', facecolor='none'
plt.title('Original Bone Image at z = {}'.format(objectSize // 2))
plt.xlabel('X')
plt.ylabel('Y')
plt.subplot(1, 2, 2)
plt.imshow(output_image, cmap='gray')
# draw little square representing the pixel on which the convolution was applied
plt.gca().add_patch(plt.Rectangle((10.5+2, 20.5+2), 1, 1, edgecolor='red', facecolor='none', lw=2))
plt.title('Output Image after Convolution')
plt.xlabel('X')
plt.ylabel('Y')

Text(0, 0.5, 'Y')
```



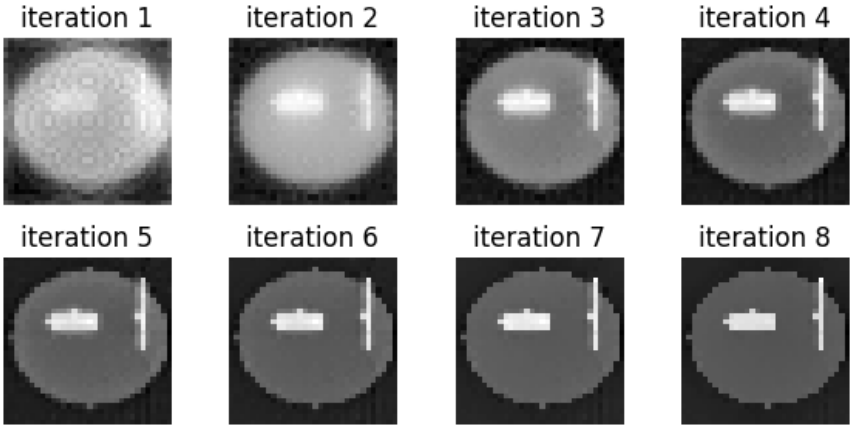
2 Attenuation coefficients water and bone

The attenuation coefficients change depending on the energy of the photons used.

```
import scipy.io
import matplotlib.pyplot as plt

# Load the .mat file
mat_data = scipy.io.loadmat('libs/materialAttenuationBoneWater.mat')

first = mat_data['materialAttenuations'][:,0]
second = mat_data['materialAttenuations'][:,1]
plt.figure(figsize=(8, 6))
plt.plot(first, label='Bone')
plt.plot(second, label='Water')
plt.xlabel('Energy (keV)')
#nicely format the unit
plt.ylabel(r"Attenuation coefficient $\mu$(cm$^{ -1}$)")
plt.legend()
plt.ylim(0,1)
plt.show()
```



```
import pickle
import matplotlib.pyplot as plt

#load iteration.pkl
with open('iteration.pkl', 'rb') as f:
    x = pickle.load(f)

# make 2x4 plot with all images in x
plt.figure(figsize=(6, 3))
for i in range(8):
    plt.subplot(2, 4, i + 1)
    plt.imshow(x[i][0, :, 8:40, 16], cmap='gray')  # e.g., bone input
    plt.title(f"iteration {i+1}")
    plt.axis('off')
plt.tight_layout()
plt.show()
```



3 The AttU-Net model

An implementation of a U-Net with attention gates added to the skip-connections

```
import numpy as np
import torch
from torch.utils.data import TensorDataset, DataLoader, random_split
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.nn import init
from torch import optim
import time
from torch.utils.data import Dataset

class conv_block(nn.Module):
    def __init__(self,ch_in,ch_out):
        super(conv_block,self).__init__()
        self.conv = nn.Sequential(
            nn.Conv3d(ch_in, ch_out, kernel_size=3,stride=1,padding=1,bias=True),
            nn.BatchNorm3d(ch_out),
            nn.ReLU(inplace=True),
            nn.Conv3d(ch_out, ch_out, kernel_size=3,stride=1,padding=1,bias=True),
            nn.BatchNorm3d(ch_out),
            nn.ReLU(inplace=True)
        )


    def forward(self,x):
        x = self.conv(x)
        return x

class up_conv(nn.Module):
    def __init__(self,ch_in,ch_out):
        super(up_conv,self).__init__()
        self.up = nn.Sequential(
            nn.Upsample(scale_factor=2),
            nn.Conv3d(ch_in,ch_out,kernel_size=3,stride=1,padding=1,bias=True),
      nn.BatchNorm3d(ch_out),
   nn.ReLU(inplace=True)
        )

    def forward(self,x):
        x = self.up(x)
        return x

class Attention_block(nn.Module):
    def __init__(self,F_g,F_l,F_int):
        super(Attention_block,self).__init__()
        self.W_g = nn.Sequential(
            nn.Conv3d(F_g, F_int, kernel_size=1,stride=1,padding=0,bias=True),
            nn.BatchNorm3d(F_int)
            )

        self.W_x = nn.Sequential(
```

```
            nn.Conv3d(F_l, F_int, kernel_size=1,stride=1,padding=0,bias=True),
            nn.BatchNorm3d(F_int)
        )

        self.psi = nn.Sequential(
            nn.Conv3d(F_int, 1, kernel_size=1,stride=1,padding=0,bias=True),
            nn.BatchNorm3d(1),
            nn.Sigmoid()
        )

        self.relu = nn.ReLU(inplace=True)

    def forward(self,g,x, return_attention=False):
        g1 = self.W_g(g)
        x1 = self.W_x(x)
        psi = self.relu(g1+x1)
        psi = self.psi(psi)
        if not return_attention:
            return x * psi
        else:
            return x * psi, psi

class AttU_Net(nn.Module):
    def __init__(self,img_ch=10,output_ch=2):
        super(AttU_Net,self).__init__()

        self.Maxpool = nn.MaxPool3d(kernel_size=2,stride=2)

        self.Conv1 = conv_block(ch_in=img_ch,ch_out=64)
        self.Conv2 = conv_block(ch_in=64,ch_out=128)
        self.Conv3 = conv_block(ch_in=128,ch_out=256)
        self.Conv4 = conv_block(ch_in=256,ch_out=512)
        self.Conv5 = conv_block(ch_in=512,ch_out=1024)

        self.Up5 = up_conv(ch_in=1024,ch_out=512)
        self.Att5 = Attention_block(F_g=512,F_l=512,F_int=256)
        self.Up_conv5 = conv_block(ch_in=1024, ch_out=512)

        self.Up4 = up_conv(ch_in=512,ch_out=256)
        self.Att4 = Attention_block(F_g=256,F_l=256,F_int=128)
        self.Up_conv4 = conv_block(ch_in=512, ch_out=256)

        self.Up3 = up_conv(ch_in=256,ch_out=128)
        self.Att3 = Attention_block(F_g=128,F_l=128,F_int=64)
        self.Up_conv3 = conv_block(ch_in=256, ch_out=128)

        self.Up2 = up_conv(ch_in=128,ch_out=64)
        self.Att2 = Attention_block(F_g=64,F_l=64,F_int=32)
        self.Up_conv2 = conv_block(ch_in=128, ch_out=64)

        self.Conv_1x1 = nn.Conv3d(64,output_ch,kernel_size=1,stride=1,padding=0)


    def forward(self,x):
```

```
        # encoding path
        x1 = self.Conv1(x)

        x2 = self.Maxpool(x1)
        x2 = self.Conv2(x2)

        x3 = self.Maxpool(x2)
        x3 = self.Conv3(x3)

        x4 = self.Maxpool(x3)
        x4 = self.Conv4(x4)

        x5 = self.Maxpool(x4)
        x5 = self.Conv5(x5)

        # decoding + concat path
        d5 = self.Up5(x5)
        x4 = self.Att5(g=d5,x=x4)
        d5 = torch.cat((x4,d5),dim=1)
        d5 = self.Up_conv5(d5)

        d4 = self.Up4(d5)
        x3 = self.Att4(g=d4,x=x3)
        d4 = torch.cat((x3,d4),dim=1)
        d4 = self.Up_conv4(d4)

        d3 = self.Up3(d4)
        x2 = self.Att3(g=d3,x=x2)
        d3 = torch.cat((x2,d3),dim=1)
        d3 = self.Up_conv3(d3)

        d2 = self.Up2(d3)
        x1 = self.Att2(g=d2,x=x1)
        d2 = torch.cat((x1,d2),dim=1)
        d2 = self.Up_conv2(d2)

        d1 = self.Conv_1x1(d2)

        return torch.sigmoid(d1)  # Apply sigmoid to the output for binary classification

    def getAttenuationMap(self, x):
        attention_maps = {}

        # Encoding path
        x1 = self.Conv1(x)
        x2 = self.Maxpool(x1)
        x2 = self.Conv2(x2)

        x3 = self.Maxpool(x2)
        x3 = self.Conv3(x3)

        x4 = self.Maxpool(x3)
        x4 = self.Conv4(x4)
```

```
        x5 = self.Maxpool(x4)
        x5 = self.Conv5(x5)

        # Decoding path with attention maps
        d5 = self.Up5(x5)
        x4, att5 = self.Att5(g=d5, x=x4, return_attention=True)
        attention_maps['att5'] = att5
        d5 = torch.cat((x4, d5), dim=1)
        d5 = self.Up_conv5(d5)

        d4 = self.Up4(d5)
        x3, att4 = self.Att4(g=d4, x=x3, return_attention=True)
        attention_maps['att4'] = att4
        d4 = torch.cat((x3, d4), dim=1)
        d4 = self.Up_conv4(d4)

        d3 = self.Up3(d4)
        x2, att3 = self.Att3(g=d3, x=x2, return_attention=True)
        attention_maps['att3'] = att3
        d3 = torch.cat((x2, d3), dim=1)
        d3 = self.Up_conv3(d3)

        d2 = self.Up2(d3)
        x1, att2 = self.Att2(g=d2, x=x1, return_attention=True)
        attention_maps['att2'] = att2
        d2 = torch.cat((x1, d2), dim=1)
        d2 = self.Up_conv2(d2)

        d1 = self.Conv_1x1(d2)
        output = torch.sigmoid(d1)

        return output, attention_maps
```

3.1 Custom dataloader

A custom dataloader has been used to load the data individually

```
class HDF5Dataset(Dataset):
    def __init__(self, h5_path, transform=None, fraction = 1.0):
        self.h5_path = h5_path
        self.transform = transform
        with h5py.File(self.h5_path, 'r') as f:
            self.length = round(f['inputs'].shape[0] * fraction)
            print(self.length)

    def __len__(self):
        return self.length

    def __getitem__(self, index):
        # Use one file per worker strategy
        with h5py.File(self.h5_path, 'r') as f:
            x = f['inputs'][index]
            y = f['outputs'][index]
```

```
        x = torch.tensor(x, dtype=torch.float32)
        y = torch.tensor(y, dtype=torch.float32)

        if self.transform:
            x = self.transform(x)
            y = self.transform(y)

        return x, y
```

3.2 Training the network

```
def train_network(network, dataloader, device, lr = 0.001, epochs=20):

    criterion = torch.nn.MSELoss()
    optimizer = torch.optim.Adam(network.parameters(), lr=lr)

    for epoch in range(epochs):
        for i, (input_batch, output_batch) in enumerate(dataloader):
            input_batch = input_batch.to(device)
            output_batch = output_batch.to(device)

            optimizer.zero_grad()
            output = network(input_batch)
            loss = criterion(output, output_batch)
            loss.backward()
            optimizer.step()

            if (i % 1000 == 0):
                open("log.txt", "a").write(f'Epoch {epoch + 1}/{epochs}, Batch {i + 1}/{len(dataloader)}
                print(f'Epoch {epoch + 1}/{epochs}, Batch {i + 1}/{len(dataloader)}, Loss: {loss.item()}
                torch.save(network.state_dict(), f'network_phantom_epoch.pth')


        open("log.txt", "a").write(f'Epoch {epoch + 1}/{epochs}, Basic Data Loss: {loss.item()}\n')
        print(f'Epoch {epoch + 1}/{epochs}, Basic Data Loss: {loss.item()}')
        torch.save(network.state_dict(), f'network_phantom.pth')


    return loss.item()  # Return the loss for the last epoch
```

4 Processing input data

- append image to sinogram
- fix dimension issues using 0 padding
- fix multiple of 16 issues using 0 padding

As input data we will append the image to the sinogram. There will be a mismatch in height which we will solve by using 0 padding. Due to the convolution layers in the network, it is desired for the input dimensions to be multiples of 16.

```
import numpy as np
from libs.simulatepreps import projectMatrix
```

4.1 Define phantom and generate projection matrix

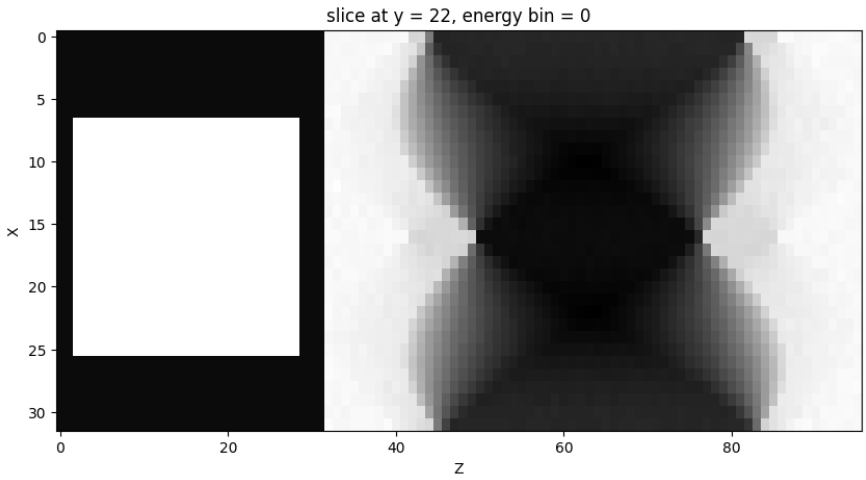
```
# generate a bone and water phantom with a 32x32 size
## SETTINGS ##
objectSize = 32                          # Object Size a square
nPixelsY = 64                            # Number of pixels on the detector plate.
nPixelsZ = 44                            # Number of pixels on the detector plate.
pixelPitch = 1                           # Pixel pitch
nPixelsPerProj = nPixelsZ * nPixelsY     # The total number of pixels on the detector plate
projections = 32                         # Number of projections


def generatePhantom(size):
    bone_cylinder = np.full((size, size, size), 0.05, dtype=np.float32)
    water_cylinder = np.full((size, size, size), 0.05, dtype=np.float32)

    radius = 0.3 * size
    center = size // 2

    for x in range(size):
        for y in range(size):
            for z in range(size):
                if ((x - center) ** 2 + (y - center) ** 2)<= radius ** 2:
                    if (z > 2 and z < size - 2):
                        bone_cylinder[x, y, z] = 1.0
                        water_cylinder[x, y, z] = 1.0

    return bone_cylinder, water_cylinder

bone, water = generatePhantom(objectSize)
x = np.column_stack((bone.ravel(), water.ravel()))
y, _, _, _ = projectMatrix(x, objectSize, nPixelsY, nPixelsZ, pixelPitch, projections)
```

4.2 define prepareInput and prepareOutput

```
def prepareInput(y):

    # Curriculum learning input preparation function
    # As an input: 10 channels consisting of empty images next to the sinograms
```

```
    # shape: 32x48x96, beginning with a 32x32x32 zero image with padding and a 32x44x64 image with padd

    zeroImage = np.zeros((objectSize, objectSize + 16, objectSize))
    zeroBin = np.zeros((projections, 2, nPixelsY))

    inputs = []

    # we have one dataset per y! with 10 channels (one for bone and one for water)
    for i, bins in enumerate(y):
        # no that is not correct

        # input is 10 channels, zeroimage appended to the energy bin
        bin = bins.reshape((projections, nPixelsZ, nPixelsY))

        # normalize so biggest value is 1, smalles value is 0
        bin = (bin - np.min(bin)) / (np.max(bin) - np.min(bin))

        bin = np.concatenate((zeroBin, bin, zeroBin), axis=1)
        # for input image, concatinate bin to zeroImage in x direction
        input_image = np.concatenate((zeroImage, bin), axis=2)
        inputs.append(input_image)
        inputs.append(input_image)

    return inputs

def prepareOutput(water, bone):

    water = np.flip(water, axis=1)  # flip along z -axis
    water = np.flip(water, axis=2)  # flip along y -axis
    bone = np.flip(bone, axis=1)    # flip along z -axis
    bone = np.flip(bone, axis=2)    # flip along y -axis

    zeroImage = np.zeros((32, 8, 32))
    zeroBin = np.zeros((projections, nPixelsZ + 4, nPixelsY))  # zero bin for the energy bin
    outputs = []

    bone_output = np.concatenate((zeroImage, bone, zeroImage), axis=1)
    water_output = np.concatenate((zeroImage, water, zeroImage), axis=1)

    bone_output = np.concatenate((bone_output, zeroBin), axis=2)
    water_output = np.concatenate((water_output, zeroBin), axis=2)

    outputs = [bone_output, water_output]  # shape will be (10, 32, 44, 64)

    return outputs

# Prepare the input and output data
input = prepareInput(y)
output = prepareOutput(water, bone)

from ipywidgets import interact, IntSlider
import matplotlib.pyplot as plt

energy_bin = 0
```

```
#invert the matrix, so z = 0 becomes z = max and y = 0 becomes y = max
flipped_bone_square = np.flip(bone, axis=1)  # flip along z -axis
flipped_bone_square = np.flip(flipped_bone_square, axis=2)  # flip along z -axis


#in input_square[0], replace [:, 8:40, :32] with the phantom
input[energy_bin][:, 8:40, :32] = flipped_bone_square[:, :, :]

# visualize with a slider, put slider for y value

plt.figure(figsize=(10, 5))
plt.imshow(input[energy_bin][:, 22, :], cmap='gray', aspect='auto')
plt.title(f'slice at y = 22, energy bin = {energy_bin}')
plt.xlabel('Z')
plt.ylabel('X')

# draw red rectangle around bottom and top padding
# draw red red rectangle from (0, 0) to (32, 8) and (0, 40) to (32, 48)
plt.show()
```



5 Visualise attention mechanism

An important part of analysing the data is the attention mechanism, which allows the model to focus on different parts of the input data. Being able to visualise this mechanism is a key aspect of understanding how the network behaves. In this notebook I will

- Load preselected data
- Run the model on the preselected data
- Visualise the attention map on the input data

Due to the size of the network, this image has been saved. Uncomment the code to reproduce.

```
from libs.network.network import AttU_Net
import pickle
import os
import torch
import matplotlib.pyplot as plt
import torch.nn.functional as F

# uncomment when running the model independently

# # Define paths
# network_path = "data/network_att_map.pth"
# data_path = "data/data_att_map.pkl"

# # load torch input tensor
# with open(data_path, "rb") as f:
#     input_tensor = pickle.load(f)


# load saved data
with open("data/attention_overlay.pkl", "rb") as f:
    data = pickle.load(f)
    input_tensor = data["input_tensor"]
    attention = data["attention"]
```

5.0.1 Load network and run output

```
# network = AttU_Net()
# network.load_state_dict(torch.load(network_path, map_location=torch.device('cpu')))  # Load the traine
# network.eval()

# output, attention_maps = network.getAttenuationMap(input_tensor)
```

5.1 Visualise attention mechanism

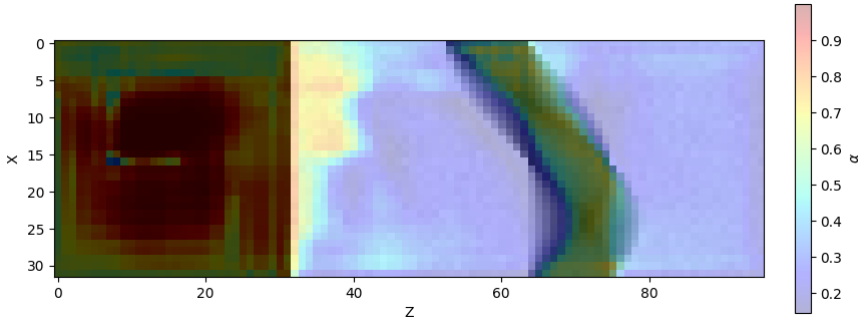
Define function show_attention_overlay which as an argument takes the input data, the attention weights, the slice axis and the slice index. For this example I have chosen to slice along the y-direction to properly visualise the sinogram.

```
def show_attention_overlay(input_tensor, attention, slice_axis=2, slice_idx=0, title="Attention Overlay
    """
    Visualize attention over an input slice.
```

```
    Args:
        input_tensor: torch.Tensor [1, C, D, H, W]
        attention:    torch.Tensor [1, 1, D', H', W']
        slice_axis:   0=D, 1=H, 2=W (which axis to slice through)
        slice_idx:    index along that axis
    """
    assert input_tensor.ndim == 5 and attention.ndim == 5

    # Resize attention to match input spatial dims
    attn_resized = F.interpolate(attention, size=input_tensor.shape[2:], mode='trilinear', align_corners

    # Select first input channel and remove batch dim
    input_tensor = input_tensor[6, 0].detach().cpu()
    attn_tensor = attn_resized[6, 0].detach().cpu()

    # Normalize attention
    attn_tensor = (attn_tensor - attn_tensor.min()) / (attn_tensor.max() - attn_tensor.min() + 1e -8)

    # Choose slice
    if slice_axis == 0:  # Depth
        input_slice = input_tensor[slice_idx, :, :].numpy()
        attn_slice  = attn_tensor[slice_idx, :, :].numpy()
    elif slice_axis == 1:  # Height
        input_slice = input_tensor[:, slice_idx, :].numpy()
        attn_slice  = attn_tensor[:, slice_idx, :].numpy()
    elif slice_axis == 2:  # Width
        input_slice = input_tensor[:, :, slice_idx].numpy()
        attn_slice  = attn_tensor[:, :, slice_idx].numpy()
    else:
        raise ValueError("slice_axis must be 0 (D), 1 (H), or 2 (W)")

    plt.figure(figsize=(10, 4))
    # Plot overlay
    plt.imshow(input_slice, cmap='gray')
    plt.imshow(attn_slice, cmap='jet', alpha=0.3)
    # make colourbar same height as image
    plt.xlabel('Z')
    plt.ylabel('X')
    plt.colorbar(label=r'$\alpha$', fraction=0.046, pad=0.04)
    plt.show()
```

5.1.1 Plot attention map

```
# attention = attention_maps['att2']
show_attention_overlay(input_tensor, attention, slice_axis=1, slice_idx=41, title="Attention from att4"
```


0
5
10
15
20
25
30
X
0
20
40
60
80
Z
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
α

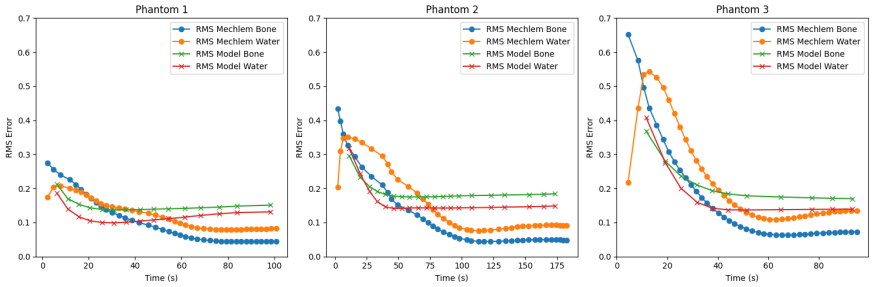
6 RMS as a function of time

A time-rms plot which includes

- rms bune itt against time
- rms water itt against time
- rms bone model against time
- rms water model against time

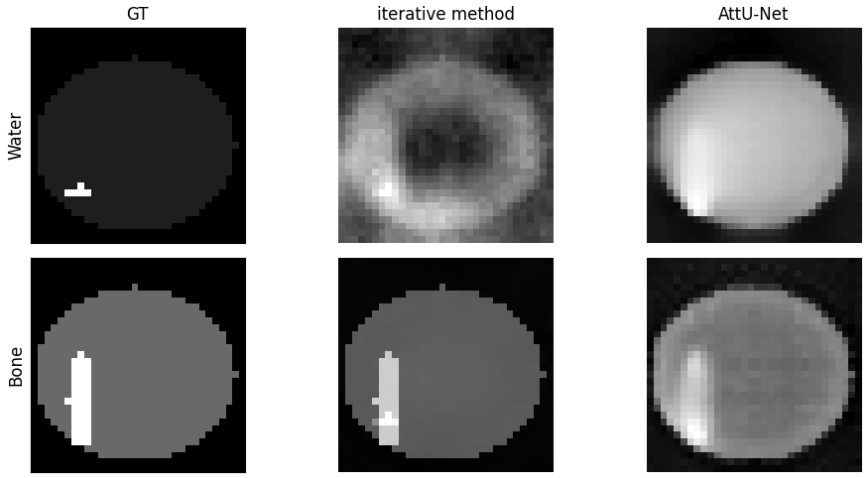
```
import pickle
import matplotlib.pyplot as plt
import numpy as np

objectSize = 32


with open('phantom_results_2.pkl', 'rb') as f:
    data = pickle.load(f)

reconstructions = data['reconstructions']
ys = data['ys']
phantoms = data['phantoms']
model_images = data['output_images']
times = data['times']
times_model = data['times_images']
rms_itt_bone = data['rms_reconstructions_bone']
rms_itt_water = data['rms_reconstructions_water']
rms_model_bone = data['rms_models_bone']
rms_model_water = data['rms_models_water']

def rms_error(image1, image2):
    if image1.shape != image2.shape:
        raise ValueError("Images must have the same dimensions")

    # Calculate the squared differences
    squared_diff = (image1 - image2) ** 2

    # Calculate the mean of the squared differences
    mean_squared_diff = np.mean(squared_diff)

    # Return the square root of the mean squared difference
    return np.sqrt(mean_squared_diff)

# calculate RMS for reconstructed images

rms_reconstructions_bone = []
rms_reconstructions_water = []
rms_models_bone = []
rms_models_water = []

for reconstruction, output_image, phantom in zip(reconstructions, model_images, phantoms):

    rms_recon_bone = []
```

```
    rms_recon_water = []
    rms_model_bone = []
    rms_model_water = []

    phantom_bone = phantom[0].transpose()  # Get the ith bone
    phantom_water = phantom[1].transpose()  # Get the ith water

    # get first image from reconstruction
    _, nMats, nIterates = reconstruction.shape
    images = reconstruction.reshape((objectSize, objectSize, objectSize, nMats, nIterates), order = 'F')

    for i in range(len(images[0, 0, 0, 0, :])):
        image_bone = images[:, :, :, 0, i]
        image_water = images[:, :, :, 1, i]
        rms_recon_bone.append(rms_error(image_bone, phantom_bone))
        rms_recon_water.append(rms_error(image_water, phantom_water))
    rms_reconstructions_bone.append(rms_recon_bone)
    rms_reconstructions_water.append(rms_recon_water)

    for image_bone, image_water in output_image:
        rms_model_bone.append(rms_error(image_bone, phantom_bone))
        rms_model_water.append(rms_error(image_water, phantom_water))
    rms_models_bone.append(rms_model_bone)
    rms_models_water.append(rms_model_water)

# we only want te keep the values for the models which are inside the time of the recon
for i in range(len(times_model)):
    times_model[i] = [t for t in times_model[i] if t <= max(times[i])]
# and the same for the RMS values
for i in range(len(rms_models_bone)):
    rms_models_bone[i] = rms_models_bone[i][:len(times_model[i])]
    rms_models_water[i] = rms_models_water[i][:len(times_model[i])]



# Plot the RMS errors for the reconstructions and the models
# make subplot with 1 row and len(reconstructions) columns
# plot them against the times
plt.figure(figsize=(15, 5))
for i in range(3):
    time_model = np.array(times_model[i])
    time_recon = np.array(times[i])
    plt.subplot(1, len(reconstructions), i + 1)
    plt.plot(time_recon, rms_reconstructions_bone[i], label='RMS Mechlem Bone', marker='o')
    plt.plot(time_recon, rms_reconstructions_water[i], label='RMS Mechlem Water', marker='o')
    plt.plot(time_model, rms_models_bone[i], label='RMS Model Bone', marker='x')
    plt.plot(time_model, rms_models_water[i], label='RMS Model Water', marker='x')
    plt.xlabel('Time (s)')
    plt.ylabel('RMS Error')
    plt.ylim([0,0.7])
    plt.title(f'Phantom {i + 1}')
    plt.legend()
plt.tight_layout()
plt.show()
```




```
# import interactive slider for displaying images
from ipywidgets import interact, IntSlider


with open('phantom_results_3.pkl', 'rb') as f:
    data = pickle.load(f)

reconstructions = data['reconstructions']
ys = data['ys']
phantoms = data['phantoms']
model_images = data['output_images']
times = data['times']
times_model = data['times_images']
rms_itt_bone = data['rms_reconstructions_bone']
rms_itt_water = data['rms_reconstructions_water']
rms_model_bone = data['rms_models_bone']
rms_model_water = data['rms_models_water']

# loop through reconstructions and display the last bone and water images
# use slider for index in reconstructions array
def display_reconstruction_images(index):
    phantom_bone = phantoms[index][0].transpose()  # Get the ith bone
    phantom_water = phantoms[index][1].transpose()  # Get the ith water


    reconstruction = reconstructions[index]
    _, nMats, nIterates = reconstruction.shape
    images = reconstruction.reshape((objectSize, objectSize, objectSize, nMats, nIterates), order='F')

    fig, ax = plt.subplots(2, 2, figsize=(10, 5))
    ax[0,0].imshow(images[:,:,objectSize // 2, 0, -1], cmap='gray')
    ax[0, 0].set_title('Phantom Bone')
    ax[0, 0].axis('off')
    ax[0, 1].imshow(images[:,:,objectSize // 2, 1, -1], cmap='gray')
    ax[0, 1].set_title('Phantom Water')
    ax[1, 0].imshow(model_images[index][2][0][:, :, objectSize // 2], cmap='gray')
    ax[1, 0].set_title('Model Bone')
    ax[1, 1].imshow(model_images[index][2][1][:, :, objectSize // 2], cmap='gray')
    ax[1, 1].set_title('Model Water')

    plt.show()

interact(display_reconstruction_images,index=IntSlider(min=0, max=len(reconstructions) -1, step=1, valu
```

```
interactive(children=(IntSlider(value=16, description='Phantom Index', max=19), Output()), _dom_classes=

<function __main__.display_reconstruction_images(index)>

# display phantom 16, make 2x2 plot with the phantom, model and iterative. one row bone one row water
phantom_index = 16
phantom_bone = phantoms[phantom_index][0].transpose()  # Get the ith bone
phantom_water = phantoms[phantom_index][1].transpose()  # Get

# the ith water
reconstruction = reconstructions[phantom_index]
_, nMats, nIterates = reconstruction.shape
images = reconstruction.reshape((objectSize, objectSize, objectSize, nMats, nIterates), order='F')
water_reconstruction = images[:, :, :, 1, -1]
bone_reconstruction = images[:, :, :, 0, -1]

model_bone = model_images[phantom_index][2][0][:, :, objectSize // 2]
model_water = model_images[phantom_index][2][1][:, :, objectSize // 2]

fig, ax = plt.subplots(2, 3, figsize=(10, 5))
ax[1, 0].imshow(phantom_bone[:, :, objectSize // 2], cmap='gray')
ax[0, 0].imshow(phantom_water[:, :, objectSize // 2], cmap='gray')
ax[0, 0].set_title('GT')
ax[1, 1].imshow(bone_reconstruction[:, :, objectSize // 2], cmap='gray')
ax[1, 1].axis('off')
ax[1, 2].imshow(model_bone, cmap='gray')
# turn of axis numbers
ax[1,2].set_xlabel('')
ax[1, 2].axis('off')
ax[0, 2].imshow(model_water, cmap='gray')
ax[0, 2].set_title('AttU -Net')
ax[0, 2].axis('off')
ax[0, 1].imshow(water_reconstruction[:, :, objectSize // 2], cmap='gray')
ax[0, 1].set_title('iterative method')
ax[0, 1].axis('off')
ax[0, 0].set_ylabel('Water', fontsize=12)
ax[1, 0].set_ylabel('Bone', fontsize=12)
ax[0, 0].set_xticks([])
ax[0, 0].set_yticks([])
ax[1, 0].set_xticks([])
ax[1, 0].set_yticks([])
plt.tight_layout()
plt.show()
```


GT
iterative method
AttU-Net
Water
Bone

7 Display validation subset

For the results we want to display 2 phantoms:

- 'good' reconstruction
- 'bad' reconstruction

We will use these reconstructions to point out the 'a model is only as good as it's data'

```
#settings
objectSize = 32     # size of the object
nPixelsY = 64       # number of pixels in Y direction detector
nPixelsZ = 44       # number of pixels in Z direction detector
projections = 64    # number of projections
iterations = 40     # number of iterations

import pickle
import matplotlib.pyplot as plt
import numpy as np

objectSize = 32


with open('phantom_results_2.pkl', 'rb') as f:
    data = pickle.load(f)

reconstructions = data['reconstructions']
ys = data['ys']
phantoms = data['phantoms']
model_images = data['output_images']
times = data['times']
times_model = data['times_images']
rms_itt_bone = data['rms_reconstructions_bone']
rms_itt_water = data['rms_reconstructions_water']
rms_model_bone = data['rms_models_bone']
rms_model_water = data['rms_models_water']

def rms_error(image1, image2):
    if image1.shape != image2.shape:
        raise ValueError("Images must have the same dimensions")

    # Calculate the squared differences
    squared_diff = (image1 - image2) ** 2

    # Calculate the mean of the squared differences
    mean_squared_diff = np.mean(squared_diff)

    # Return the square root of the mean squared difference
    return np.sqrt(mean_squared_diff)

def get_best_image(reconstruction, bone, water):

    # calculate RMS error for each image, return best bone, water pair
    best_rms = float('inf')
```

```
    best_bone = None
    best_water = None

    _, nMats, nIterates = reconstruction.shape
    images = reconstruction.reshape((objectSize, objectSize, objectSize, nMats, nIterates), order = 'F'

    for i in range(nIterates):
        bone_reconstructed = images[:, :, :, 0, i]  # Maybe bone
        water_reconstructed = images[:, :, :, 1, i]  # Maybe water

        # Calculate RMS error
        rms_bone = rms_error(bone_reconstructed, bone)
        rms_water = rms_error(water_reconstructed, water)
        total_rms = rms_bone + rms_water
        if total_rms < best_rms:
            best_rms = total_rms
            best_bone = bone_reconstructed
            best_water = water_reconstructed

    return best_bone, best_water, best_rms
```

7.1 display reconstructions and GT

```
# import interact, display for each index (as a slider) the final model_image, recon image and phantom
from ipywidgets import interact, FloatSlider, IntSlider, fixed

def display_images(index, itt):

    # get GT for water
    phantom = phantoms[index]  # Get the ith phantom
    phantom_water = phantom[1].transpose()  # Get the water phantom
    phantom_bone = phantom[0].transpose()  # Get the bone phantom

    # get best iterative image
    reconstruction = reconstructions[index]  # Get the ith reconstruction
    best_bone_image , best_water_image, rms = get_best_image(reconstruction, phantom_bone, phantom_wate

    # last iterative image
    _, nMats, nIterates = reconstruction.shape
    images = reconstruction.reshape((objectSize, objectSize, objectSize, nMats, nIterates), order = 'F'
    last_water_image = images[:, :, :, 1, -1]  # Get the last water image
    last_bone_image = images[:, :, :, 0, -1]  # Get the last bone image

    model_image = model_images[index][itt]  # Get the ith output image
    model_bone = model_image[0]  # Get the bone part of the model imag
    model_water = model_image[1]  # Get the water part of the model image

    # display ONLY water images
    plt.figure(figsize=(7, 4))
    plt.subplot(2, 4, 1)
    plt.imshow(phantom_water[:,:,objectSize//2], cmap='gray')
    plt.title(' (a)')
    plt.axis('off')
    plt.subplot(2, 4, 2)
```

```
    plt.imshow(best_water_image[:,:,objectSize//2], cmap='gray')
    plt.title(f' (b) ')
    plt.axis('off')
    plt.subplot(2, 4, 3)
    plt.imshow(last_water_image[:,:,objectSize//2], cmap='gray')
    plt.title(' (c) ')
    plt.axis('off')
    plt.subplot(2, 4, 4)
    plt.imshow(model_water[:,:,objectSize//2], cmap='gray')
    plt.title(f' (d) ')
    plt.axis('off')
    plt.subplot(2, 4, 5)
    plt.imshow(phantom_bone[:,:,objectSize//2], cmap='gray')
    plt.title(' (e)')
    plt.axis('off')
    plt.subplot(2, 4, 6)
    plt.imshow(best_bone_image[:,:,objectSize//2], cmap='gray')
    plt.title(f' (f) ')
    plt.axis('off')
    plt.subplot(2, 4, 7)
    plt.imshow(last_bone_image[:,:,objectSize//2], cmap='gray')
    plt.title(' (g) ')
    plt.axis('off')
    plt.subplot(2, 4, 8)
    plt.imshow(model_bone[:,:,objectSize//2], cmap='gray')
    plt.title(f' (h) ')
    plt.axis('off')
```

```
interact(display_images, index=IntSlider(min=0, max=len(model_images) -1, step=1, value=0, description=
```

```
interactive(children=(IntSlider(value=0, description='Index', max=2), IntSlider(value=0, description='I
```

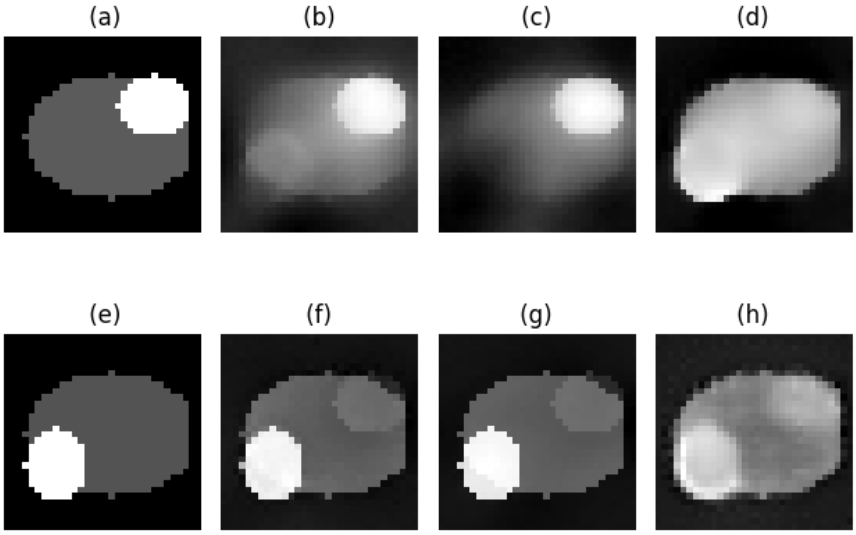
```
<function __main__.display_images(index, itt)>
```

7.2 Image to display

choose to display phantom 1 at iteration 3

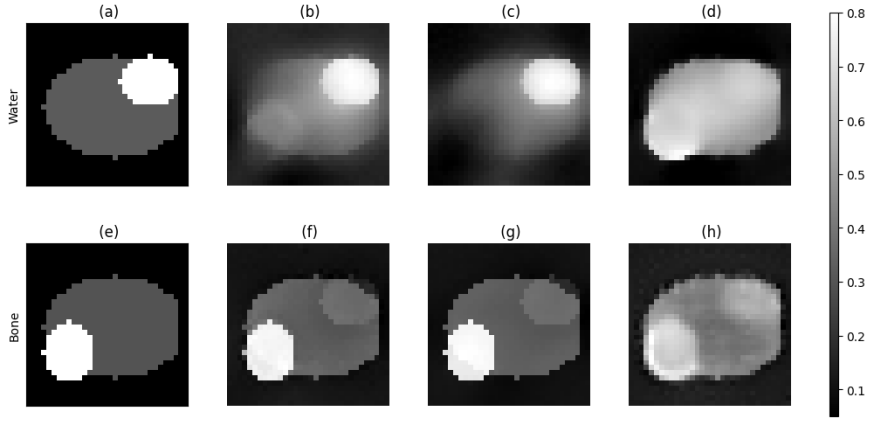
```
index = 1
itt = 3

# get GT for water
phantom = phantoms[index]  # Get the ith phantom
phantom_water = phantom[1].transpose()  # Get the water phantom
phantom_bone = phantom[0].transpose()  # Get the bone phantom

# get best iterative image
reconstruction = reconstructions[index]  # Get the ith reconstruction
best_bone_image , best_water_image, rms = get_best_image(reconstruction, phantom_bone, phantom_water)

# last iterative image
```

```
_, nMats, nIterates = reconstruction.shape
images = reconstruction.reshape((objectSize, objectSize, objectSize, nMats, nIterates), order = 'F')
last_water_image = images[:, :, :, 1, -1]  # Get the last water image
last_bone_image = images[:, :, :, 0, -1]  # Get the last bone image

model_image = model_images[index][itt]  # Get the ith output image
model_bone = model_image[0]  # Get the bone part of the model imag
model_water = model_image[1]  # Get the water part of the model image

# display ONLY water images
# get constant colorbar for all images
plt.subplot(2, 4, 1)
plt.imshow(phantom_water[:,:,objectSize//2], cmap='gray')
plt.title(' (a)')
plt.axis('off')
plt.subplot(2, 4, 2)
plt.imshow(best_water_image[:,:,objectSize//2], cmap='gray')
plt.title(f' (b) ')
plt.axis('off')
plt.subplot(2, 4, 3)
plt.imshow(last_water_image[:,:,objectSize//2], cmap='gray')
plt.title(' (c) ')
plt.axis('off')
plt.subplot(2, 4, 4)
plt.imshow(model_water[:,:,objectSize//2], cmap='gray')
plt.title(f' (d) ')
plt.axis('off')
plt.subplot(2, 4, 5)
plt.imshow(phantom_bone[:,:,objectSize//2], cmap='gray')
plt.title(' (e)')
plt.axis('off')
plt.subplot(2, 4, 6)
plt.imshow(best_bone_image[:,:,objectSize//2], cmap='gray')
plt.title(f' (f) ')
plt.axis('off')
plt.subplot(2, 4, 7)
plt.imshow(last_bone_image[:,:,objectSize//2], cmap='gray')
plt.title(' (g) ')
plt.axis('off')
plt.subplot(2, 4, 8)
plt.imshow(model_bone[:,:,objectSize//2], cmap='gray')
plt.title(f' (h) ')
plt.axis('off')
plt.tight_layout()
plt.show()
```



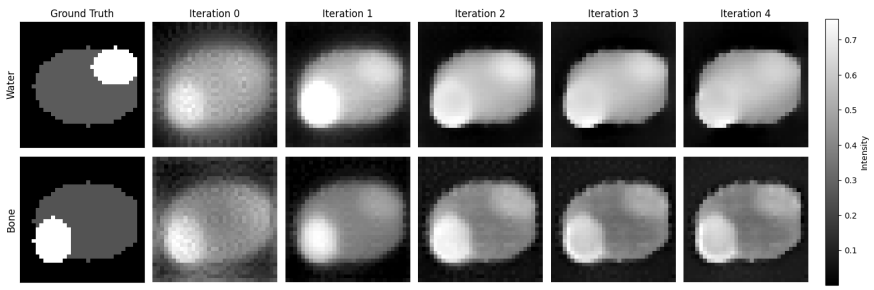

```
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec

fig = plt.figure(figsize=(12, 6))
gs = gridspec.GridSpec(2, 5, width_ratios=[1, 1, 1, 1, 0.05], wspace=0.3)

# First row
ax1 = fig.add_subplot(gs[0, 0])
im = ax1.imshow(phantom_water[:, :, objectSize//2], cmap='gray')
ax1.set_title(' (a)')
ax1.set_xticks([])
ax1.set_yticks([])
ax1.set_ylabel('Water')

ax2 = fig.add_subplot(gs[0, 1])
ax2.imshow(best_water_image[:, :, objectSize//2], cmap='gray')
ax2.set_title(' (b)')
ax2.axis('off')

ax3 = fig.add_subplot(gs[0, 2])
ax3.imshow(last_water_image[:, :, objectSize//2], cmap='gray')
ax3.set_title(' (c)')
ax3.axis('off')

ax4 = fig.add_subplot(gs[0, 3])
ax4.imshow(model_water[:, :, objectSize//2], cmap='gray')
ax4.set_title(' (d)')
ax4.axis('off')

# Second row
ax5 = fig.add_subplot(gs[1, 0])
ax5.imshow(phantom_bone[:, :, objectSize//2], cmap='gray')
ax5.set_title(' (e)')
ax5.set_xticks([])
ax5.set_yticks([])
ax5.set_ylabel('Bone')
```

```
ax6 = fig.add_subplot(gs[1, 1])
ax6.imshow(best_bone_image[:, :, objectSize//2], cmap='gray')
ax6.set_title(' (f)')
ax6.axis('off')

ax7 = fig.add_subplot(gs[1, 2])
ax7.imshow(last_bone_image[:, :, objectSize//2], cmap='gray')
ax7.set_title(' (g)')
ax7.axis('off')

ax8 = fig.add_subplot(gs[1, 3])
ax8.imshow(model_bone[:, :, objectSize//2], cmap='gray')
ax8.set_title(' (h)')
ax8.axis('off')

# Add colorbar on dedicated axis (right column)
cax = fig.add_subplot(gs[:, 4])
fig.colorbar(im, cax=cax)

# fig.tight_layout()
plt.show()
```



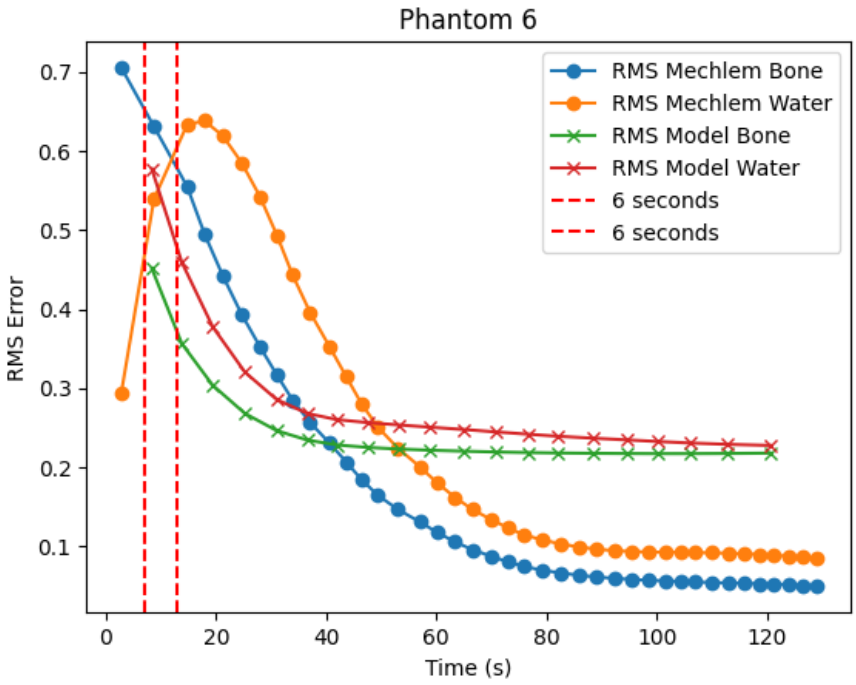

```
import matplotlib.pyplot as plt
import numpy as np

images = model_images[1]
reconstruction = reconstructions[index]  # Get the ith reconstruction
images_r = reconstruction.reshape((objectSize, objectSize, objectSize, 2, -1), order='F')  # Reshape to

first_water = images_r[:, :, :, 1, 0]  # Get the last water image
first_bone = images_r[:, :, :, 0, 0]  # Get the last bone image

len_images = 5

fig, axes = plt.subplots(2, len_images+1, figsize=(15, 5))
# We'll store the last image displayed for colorbar reference
im = None

for i in range(len_images -1):
```

```
    # Top row (second index 1)
    im = axes[0, i+2].imshow(images[i][1][:, :, objectSize // 2], cmap='gray')
    axes[0, i+2].set_title(f'Iteration {i+1}')
    axes[0, i+2].axis('off')

    # Bottom row (first index 0)
    axes[1, i+2].imshow(images[i][0][:, :, objectSize // 2], cmap='gray')
    axes[1, i+2].axis('off')

# add iteration 0 images
axes[0, 1].imshow(first_water[:, :, objectSize // 2], cmap='gray')
axes[0, 1].set_title('Iteration 0')
axes[1, 1].imshow(first_bone[:, :, objectSize // 2], cmap='gray')
# Add titles for the first column
axes[0, 1].axis('off')
axes[1, 1].axis('off')

# ground truth at [0, 0] and [1, 0]
axes[0, 0].imshow(phantoms[1][1].transpose()[:, :, objectSize // 2], cmap='gray')
axes[0, 0].set_title('Ground Truth')
axes[1, 0].imshow(phantoms[1][0].transpose()[:, :, objectSize // 2], cmap='gray')
# axes[0, 0].axis('off')
# axes[1, 0].axis('off')

# add general label to y -axis of each row, of the axis in general
axes[0, 0].set_ylabel('Water', fontsize=12)
axes[1, 0].set_ylabel('Bone', fontsize=12)

# turn off x -axis labels for all but the last column
for ax in axes[0, : -1]:
    ax.set_xticks([])
    ax.set_yticks([])
for ax in axes[1, : -1]:
    ax.set_xticks([])
    ax.set_yticks([])


plt.tight_layout()

# Add a single colorbar on the right, aligned to the height of the figure
cbar = fig.colorbar(im, ax=axes, orientation='vertical', fraction=0.05, pad=0.02)
cbar.set_label('Intensity')

plt.show()
```



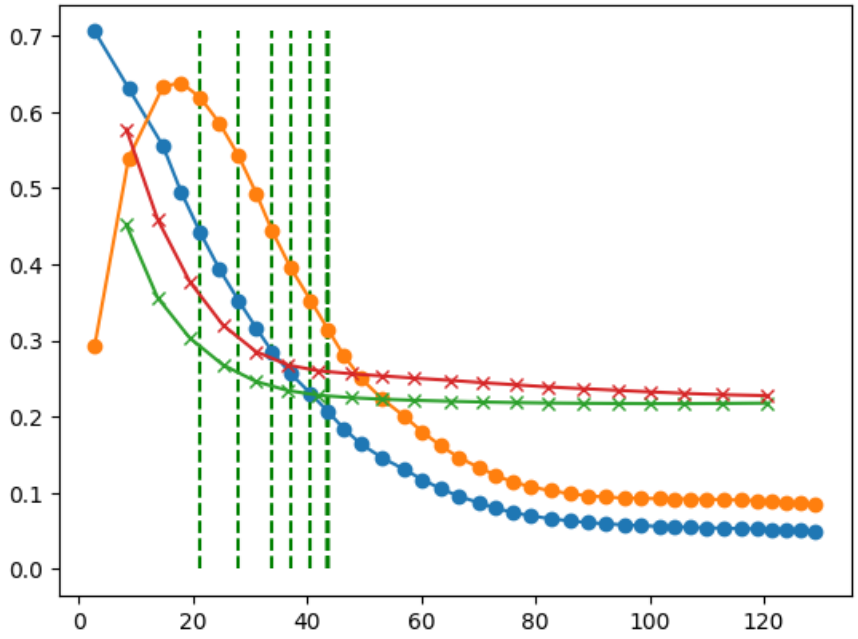
```
import pickle
import numpy as np
import matplotlib.pyplot as plt

## SETTINGS ##
objectSize = 32                         # Object Size a square
nPixelsY = 64                           # Number of pixels on the detector plate.
nPixelsZ = 44                           # Number of pixels on the detector plate.
pixelPitch = 1                          # Pixel pitch
nPixelsPerProj = nPixelsZ * nPixelsY    # The total number of pixels on the detector plate
projections = 32                        # Number of projections
nSubset = 1
nIter = 40

data_file = "phantom_results_3.pkl"

def rms_error(image1, image2):
    if image1.shape != image2.shape:
        raise ValueError("Images must have the same dimensions")

    # Calculate the squared differences
    squared_diff = (image1 - image2) ** 2

    # Calculate the mean of the squared differences
    mean_squared_diff = np.mean(squared_diff)

    # Return the square root of the mean squared difference
    return np.sqrt(mean_squared_diff)

# import pickle data
with open(data_file, 'rb') as f:
    data = pickle.load(f)

print(data.keys())
ys = data['ys']
times = data['times']
reconstructions = data['reconstructions']
phantoms = data['phantoms']
model_images = data['output_images']
model_times = data['times_images']

dict_keys(['reconstructions', 'ys', 'times', 'phantoms', 'output_images', 'times_images', 'rms_reconstru

# calculate RMS for reconstructed images

rms_reconstructions_bone = []
rms_reconstructions_water = []
rms_models_bone = []
rms_models_water = []

for reconstruction, output_image, phantom in zip(reconstructions, model_images, phantoms):

    rms_recon_bone = []
    rms_recon_water = []
    rms_model_bone = []
```

```
    rms_model_water = []

    phantom_bone = phantom[0].transpose()  # Get the ith bone
    phantom_water = phantom[1].transpose()  # Get the ith water

    # get first image from reconstruction
    _, nMats, nIterates = reconstruction.shape
    images = reconstruction.reshape((objectSize, objectSize, objectSize, nMats, nIterates), order = 'F')

    for i in range(len(images[0, 0, 0, 0, :])):
        image_bone = images[:, :, :, 0, i]
        image_water = images[:, :, :, 1, i]
        rms_recon_bone.append(rms_error(image_bone, phantom_bone))
        rms_recon_water.append(rms_error(image_water, phantom_water))
    rms_reconstructions_bone.append(rms_recon_bone)
    rms_reconstructions_water.append(rms_recon_water)

    for image_bone, image_water in output_image:
        rms_model_bone.append(rms_error(image_bone, phantom_bone))
        rms_model_water.append(rms_error(image_water, phantom_water))
    rms_models_bone.append(rms_model_bone)
    rms_models_water.append(rms_model_water)

i = 5
time_model = np.array(model_times[i])
time_recon = np.array(times[i])

plt.plot(time_recon, rms_reconstructions_bone[i], label='RMS Mechlem Bone', marker='o')
plt.plot(time_recon, rms_reconstructions_water[i], label='RMS Mechlem Water', marker='o')
plt.plot(time_model, rms_models_bone[i], label='RMS Model Bone', marker='x')
plt.plot(time_model, rms_models_water[i], label='RMS Model Water', marker='x')
# vertical bar at 6
plt.axvline(x=7, color='r', linestyle=' - -', label='6 seconds')
plt.axvline(x=12.9, color='r', linestyle=' - -', label='6 seconds')

plt.xlabel('Time (s)')
plt.ylabel('RMS Error')
plt.title(f'Phantom {i + 1}')
plt.legend()
plt.show()
```

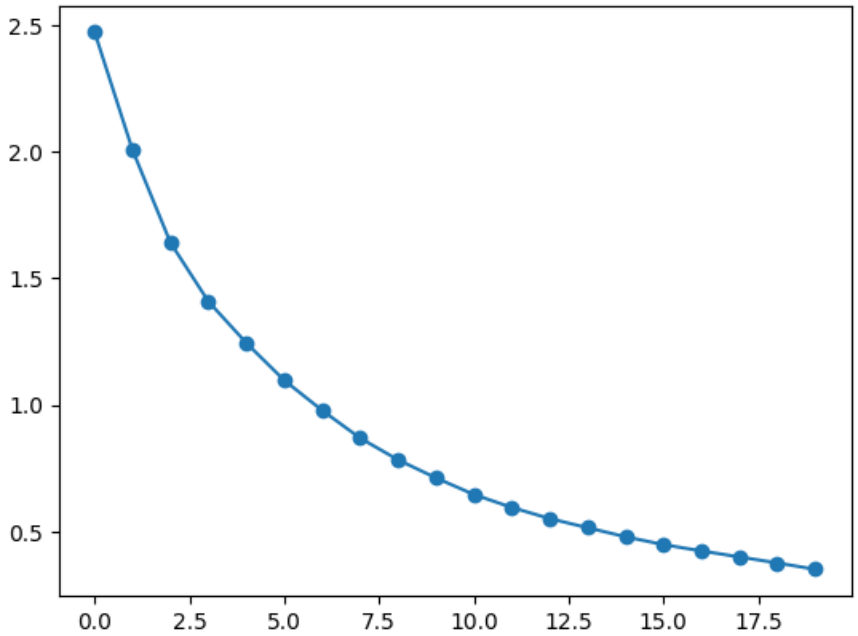



```
first_times = [t[0] for t in model_times]
print("first times:", first_times)
print("average first time:", np.mean(first_times), "+ -", np.std(first_times), "seconds")

speedups = []
interpolated_speedups = []

for i in range(len(rms_reconstructions_bone)):
    rms_bone_model = rms_models_bone[i][0]
    rms_water_model = rms_models_water[i][0]

    for j in range(len(rms_reconstructions_bone[i])):

        if(rms_reconstructions_bone[i][j] < rms_bone_model):

            speedup = times[i][j] / first_times[i]
            speedups.append(speedup)

            # also interpolate to see at what time the rms is the same
            slope = (rms_reconstructions_bone[i][j+1] - rms_reconstructions_bone[i][j]) / (times[i][j+1]

            # see for what time the rms is the same
            if slope != 0:
                interpolated_time = (rms_bone_model - rms_reconstructions_bone[i][j]) / slope + times[i]
                interpolated_speedups.append(interpolated_time / first_times[i])
            else:
                interpolated_speedups.append(speedup)
            break


print("average speedup:", np.mean(speedups), "+ -", np.std(speedups), "after", np.mean(first_times), "+
```

```
print("average interpolated speedup:", np.mean(interpolated_speedups), "+ -", np.std(interpolated_speedu

first times: [6.995673656463623, 8.126861333847046, 9.385169744491577, 13.482990264892578, 8.5782411098
average first time: 9.347357761859893 + - 1.5048523200384376 seconds
average speedup: 1.6119690273535912 + - 0.3327647687994946 after 9.347357761859893 + - 1.50485232003843
average interpolated speedup: 1.4428355769342633 + - 0.3332045100786529 after 9.347357761859893 + - 1.5

itt = 5

time_model = model_times[itt]
time_recon = times[itt]

rms_model_bone = rms_models_bone[itt]
rms_model_water = rms_models_water[itt]
rms_recon_bone = rms_reconstructions_bone[itt]
rms_recon_water = rms_reconstructions_water[itt]

interpolated_speedup = []

for idx in range(len(rms_model_bone)):

    rms_bone = rms_model_bone[idx]
    rms_water = rms_model_water[idx]

    # print(f"RMS Model Bone: {rms_bone:.4f}, RMS Model Water: {rms_water:.4f}")
    for i in range(len(rms_recon_bone)):
        if(rms_recon_bone[i] < rms_bone):
            plt.vlines(x=time_recon[i], ymin=0, ymax=max(rms_recon_bone + rms_recon_water), color='g', l

            # interpolate to get speedup
            slope = (rms_recon_bone[i+1] - rms_recon_bone[i]) / (time_recon[i+1] - time_recon[i])
            if slope != 0:
                interpolated_time = (rms_bone - rms_recon_bone[i]) / slope + time_recon[i]
                interpolated_speedup.append(interpolated_time / time_model[idx])
            else:
                interpolated_speedup.append(time_recon[i] / time_model[idx])
            break
        pass

plt.plot(time_recon, rms_recon_bone, label='RMS Mechlem Bone', marker='o')
plt.plot(time_recon, rms_recon_water, label='RMS Mechlem Water', marker='o')
plt.plot(time_model, rms_model_bone, label='RMS Model Bone', marker='x')
plt.plot(time_model, rms_model_water, label='RMS Model Water', marker='x')
plt.show()

plt.plot(interpolated_speedup, label='Interpolated Speedup', marker='o')
```



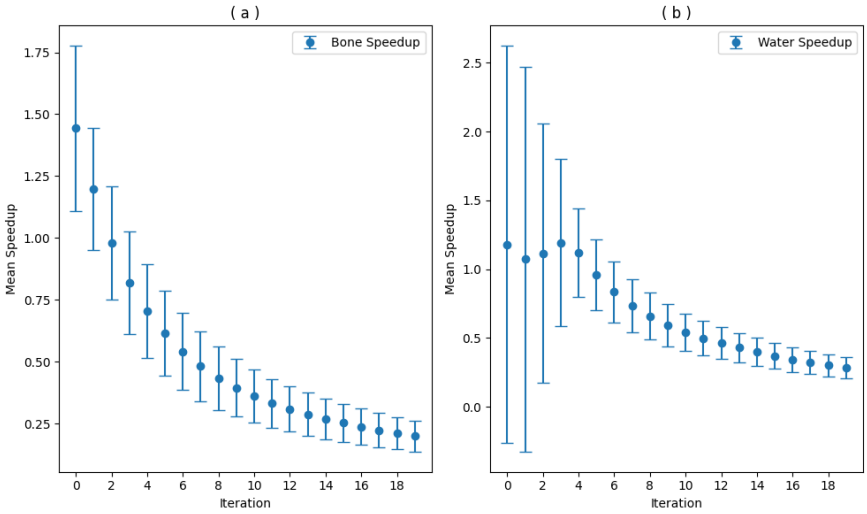

```
[<matplotlib.lines.Line2D at 0x17d8427b460>]
```




```
all_interpolated_speedups_bone = []  # List of lists - one list per phantom
all_interpolated_speedups_water = []  # List of lists - one list per phantom

for itt in range(len(times)):  # Loop over phantoms
    time_model = model_times[itt]
    time_recon = times[itt]

    rms_model_bone = rms_models_bone[itt]
    rms_model_water = rms_models_water[itt]
```

```
    rms_recon_bone = rms_reconstructions_bone[itt]
    rms_recon_water = rms_reconstructions_water[itt]

    interpolated_speedup_bone = []
    interpolated_speedup_water = []

    for idx in range(len(rms_model_bone)):  # Loop over model iterations (idx)
        rms_bone = rms_model_bone[idx]
        rms_water = rms_model_water[idx]

        for i in range(len(rms_recon_bone) - 1):  # Loop over recon steps
            if rms_recon_bone[i] < rms_bone:
                # interpolate to get speedup
                slope = (rms_recon_bone[i+1] - rms_recon_bone[i]) / (time_recon[i+1] - time_recon[i])
                if slope != 0:
                    interpolated_time = (rms_bone - rms_recon_bone[i]) / slope + time_recon[i]
                    interpolated_speedup_bone.append(interpolated_time / time_model[idx])
                else:
                    interpolated_speedup_bone.append(time_recon[i] / time_model[idx])
                break  # Once found, move to next model iteration

        for i in range(len(rms_recon_water) - 1):  # Loop over recon steps for water
            if rms_recon_water[i] < rms_water:
                # interpolate to get speedup
                slope = (rms_recon_water[i+1] - rms_recon_water[i]) / (time_recon[i+1] - time_recon[i])
                if slope != 0:
                    interpolated_time = (rms_water - rms_recon_water[i]) / slope + time_recon[i]
                    interpolated_speedup_water.append(interpolated_time / time_model[idx])
                else:
                    interpolated_speedup_water.append(time_recon[i] / time_model[idx])
                break

    all_interpolated_speedups_water.append(interpolated_speedup_water)  # Save speedup curve for this pl
    all_interpolated_speedups_bone.append(interpolated_speedup_bone)

max_len_bone  = max(len(x) for x in all_interpolated_speedups_bone)
max_len_water = max(len(x) for x in all_interpolated_speedups_water)

bone_array  = np.array([x + [np.nan] * (max_len_bone  - len(x)) for x in all_interpolated_speedups_bone]
water_array = np.array([x + [np.nan] * (max_len_water - len(x)) for x in all_interpolated_speedups_wate

mean_speedups_bone = np.nanmean(bone_array, axis=0)
std_speedups_bone  = np.nanstd(bone_array, axis=0)

mean_speedups_water = np.nanmean(water_array, axis=0)
std_speedups_water  = np.nanstd(water_array, axis=0)

plt.figure(figsize=(10, 6))
plt.subplot(1, 2, 1)
plt.errorbar(range(len(mean_speedups_bone)), mean_speedups_bone, yerr
=std_speedups_bone, fmt='o', label='Bone Speedup', capsize=5)
plt.xlabel('Iteration')
plt.ylabel('Mean Speedup')
plt.title('( a )')
```

```
plt.xticks(range(0,len(mean_speedups_bone), 2))
plt.legend()
plt.subplot(1, 2, 2)
plt.errorbar(range(len(mean_speedups_water)), mean_speedups_water, yerr=std_speedups_water, fmt='o', lab
plt.xlabel('Iteration')
plt.ylabel('Mean Speedup')
plt.title('( b )')
plt.legend()
plt.xticks(range(0,len(mean_speedups_bone),2))
plt.tight_layout()
plt.show()
```




```
# we want to calculate the average time taken to reach a certain error for both algorithms
# rms_reconstructions_bone = []
# rms_reconstructions_water = []
# rms_models_bone = []
# rms_models_water = []

rms_errors = [0.5, 0.4, 0.3, 0.2, 0.1]  # Define the RMS error thresholds

# we want to calculate the average time the squared sum of the rms is below the error
for error in rms_errors:
    times_to_reach_error = []
    for i in range(len(rms_reconstructions_bone)):
        for j in range(len(rms_reconstructions_bone[i])):
            if np.sqrt(rms_reconstructions_bone[i][j]**2) < error:
                times_to_reach_error.append(times[i][j])
                break  # Stop at the first occurrence of the error threshold

    if times_to_reach_error:
        average_time = np.mean(times_to_reach_error)
        print(f"Average time to reach RMS error {error}: {average_time:.2f} seconds")
        plt.plot(average_time, error, 'o', label=f'Mechlem {error} seconds')

for error in rms_errors:
    times_to_reach_error = []
    for i in range(len(rms_models_bone)):
```

```
        for j in range(len(rms_models_bone[i])):
            if np.sqrt(rms_models_bone[i][j]**2) < error:
                times_to_reach_error.append(model_times[i][j])
                break  # Stop at the first occurrence of the error threshold
    if times_to_reach_error:
        average_time = np.mean(times_to_reach_error)
        print(f"Average time to reach RMS error {error} for model: {average_time:.2f} seconds")
        plt.plot(average_time, error, 'o', label=f'Model {error} seconds')
```

```
Average time to reach RMS error 0.5: 4.75 seconds
Average time to reach RMS error 0.4: 7.38 seconds
Average time to reach RMS error 0.3: 13.58 seconds
Average time to reach RMS error 0.2: 23.68 seconds
Average time to reach RMS error 0.1: 42.03 seconds
Average time to reach RMS error 0.5 for model: 9.35 seconds
Average time to reach RMS error 0.4 for model: 10.48 seconds
Average time to reach RMS error 0.3 for model: 14.09 seconds
Average time to reach RMS error 0.2 for model: 23.00 seconds
Average time to reach RMS error 0.1 for model: 200.13 seconds
```